

Borel measurable factor-subspace selection in parametrized weakly compact factorization

arXiv automatic math run `fa_banach_001`

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Source Question

This packet addresses Question 6.1 of Antunes–Beanland–Braga, *Uniformly Factoring Weakly Compact operators and Parametrized Dualization*, arXiv:1909.07475 [1]. The question appears on PDF page 30.

Source: [arXiv:1909.07475v1 \[math.FA\]](#)

6. CONCLUDING REMARKS AND QUESTIONS

We finish this paper with some natural questions which are left open. Firstly, it would be very interesting to get rid of $\sigma(\Sigma_1^1)$ -measurability in the theorems in Subsection 5.2. Since an arbitrary analytic subset of $\mathbb{A} \subset \mathbb{N}^{\mathbb{N}} \times \mathbb{N}^{\mathbb{N}}$ may not have a Borel uniformization even if $p(\mathbb{A}) = \mathbb{N}^{\mathbb{N}}$, a proof for that will probably require some new ideas and a different amalgamation method.

Question 6.1. Do the Borel measurable versions of Theorem 5.1, Corollary 5.12 and Corollary 5.13 hold?

As noticed in the introduction, if $\mathcal{C}$ is a collection of weakly compact operators containing the identity on a Banach space without the bounded

The question asks whether the Borel measurable versions of Theorem 5.1, Corollary 5.12, and Corollary 5.13 of the source paper hold. The answer is yes.

Proof Intuition

The proof of Theorem 5.1 in [1] already constructs Borel maps up to the final rational-amalgamation step. At that point the paper has a Borel code map

$$\eta = \psi \circ \sigma : \mathcal{B} \rightarrow \mathbb{N}^{\mathbb{N}}$$

whose range is analytic. The published proof chooses an arbitrary closed tree whose projection is this analytic range and then invokes Jankov–von Neumann uniformization, obtaining only $\sigma(\Sigma_1^1)$ -measurability.

The key observation is that this arbitrary tree is unnecessary. Since the analytic set in question is specifically the image of a Borel map, one can build the closed tree from a closed parametrization of the Borel graph of η . This gives a Borel branch selector for the actual parameter $A \in \mathcal{B}$. Every branch of the new tree still has first coordinate $\eta(A_0)$ for some original $A_0 \in \mathcal{B}$, so the reflexivity argument in the Kurka amalgamation step is unchanged.

A closed-tree lift for Borel images

Lemma 1. *Let S be a standard Borel space and let $\eta : S \rightarrow \mathbb{N}^{\mathbb{N}}$ be Borel. Then there are a closed set $C \subset \mathbb{N}^{\mathbb{N}} \times \mathbb{N}^{\mathbb{N}}$ and a Borel map*

$$\tau : S \rightarrow C$$

such that, writing p for projection onto the first coordinate,

$$p(C) = \eta(S) \quad \text{and} \quad p(\tau(s)) = \eta(s) \quad (s \in S).$$

Moreover every branch $(\alpha, \gamma) \in C$ has $\alpha = \eta(s)$ for some $s \in S$.

Proof. Choose a Borel isomorphism $i : S \rightarrow S_0$, where S_0 is a Borel subset of $\mathbb{N}^{\mathbb{N}}$. The map

$$s \mapsto (i(s), \eta(s))$$

is Borel and injective, so its image

$$G = \{(i(s), \eta(s)) : s \in S\} \subset \mathbb{N}^{\mathbb{N}} \times \mathbb{N}^{\mathbb{N}}$$

is Borel by the Lusin–Souslin theorem.

Use the standard representation theorem that every Borel subset of a Polish space is a one-to-one continuous image of a closed subset of Baire space [3, Theorem 13.7]. Thus there are a closed set $P \subset \mathbb{N}^{\mathbb{N}}$ and a continuous bijection

$$h : P \rightarrow G.$$

Write $h(z) = (h_1(z), h_2(z))$. Define

$$C = \{(h_2(z), z) : z \in P\}.$$

Since P is closed and h_2 is continuous, C is closed. Also $p(C) = h_2(P) = \eta(S)$.

The inverse $h^{-1} : G \rightarrow P$ is Borel, again by Lusin–Souslin. Hence

$$\tau(s) = (\eta(s), h^{-1}(i(s), \eta(s)))$$

is a Borel map from S into C , and $p(\tau(s)) = \eta(s)$. The final assertion follows from $p(C) = \eta(S)$. $\square$

Borel version of Theorem 5.1

Theorem 1. *Let $\mathcal{B} \subset \mathcal{W}_{\text{SB}, \text{SD}}$ be Borel. Then the maps in Theorem 5.1 of [1] may be chosen Borel: there are a reflexive space $Z \in \text{SB}$, a Borel map $\Psi : \mathcal{B} \rightarrow \text{SB}(Z)$, and a Borel map*

$$\Phi : \mathcal{D} \rightarrow Z, \quad \mathcal{D} = \{(X, Y, \widehat{A}, x) \in \mathcal{B} \times C(\Delta) : x \in X\},$$

such that for every $A = (X, Y, \widehat{A}) \in \mathcal{B}$,

1. $\Phi_A(X) = \Psi(A)$,
2. $\Phi_A : X \rightarrow Z$ is bounded linear with $\|\Phi_A\| \leq 1$, and
3. $A = L \circ \Phi_A$ for some bounded operator $L : \Psi(A) \rightarrow Y$.

Proof. We use the notation of the proof of Theorem 5.1 in [1]. The preceding factorization-through-a-Borel-basic-sequence theorem in that paper gives Borel maps

$$\sigma : \mathcal{B} \rightarrow C(\Delta)^{\mathbb{N}}, \quad \varphi : \mathcal{D} \rightarrow C(\Delta),$$

where $\sigma(A)$ is a normalized shrinking boundedly complete basic sequence and A factors through $\overline{\text{span}}\{\sigma(A)\}$ via φ_A . Let $\psi : \text{mbs} \rightarrow \mathbb{N}^{\mathbb{N}}$ be the rational-code map from Lemma 5.10 of the source paper, and set

$$\eta(A) = \psi(\sigma(A)).$$

Then $\eta : \mathcal{B} \rightarrow \mathbb{N}^{\mathbb{N}}$ is Borel.

Apply Lemma 1 to $S = \mathcal{B}$ and this η . Let $C \subset \mathbb{N}^{\mathbb{N}} \times \mathbb{N}^{\mathbb{N}}$ be the closed set obtained there, and let $\tau : \mathcal{B} \rightarrow C$ be the Borel lift. Let T be the pruned tree on $\mathbb{N} \times \mathbb{N}$ with $[T] = C$.

Now repeat the Kurka amalgamation construction from the proof of Theorem 5.1, using this tree T . Namely define the norm on $c_{00}(T)$ by

$$\|x\| = \sup_{\beta \in [T]} \left\| \sum_{t \prec \beta} x(t) r_{|t|} \right\|_{p(\beta)},$$

let E be the completion, let E_{β} be the branch subspace, set

$$W = \overline{\text{conv}} \bigcup_{\beta \in [T]} B_{E_{\beta}}, \quad Z = \Delta_2(E, W),$$

and view Z as an element of SB.

The reflexivity proof is unchanged. Indeed, if $\beta \in [T] = C$, then $p(\beta) = \eta(A_0) = \psi(\sigma(A_0))$ for some $A_0 \in \mathcal{B}$ by Lemma 1. The source proof shows that $F_{\psi(\sigma(A_0))}$ is reflexive. Hence every branch space E_{β} is reflexive, and Kurka's proposition used in the source proof implies that Z is reflexive.

For $A \in \mathcal{B}$, put $t_A = \tau(A)$. Since $p(t_A) = \eta(A)$, the same branch isometry as in the source proof embeds $F_{\eta(A)}$ into Z along the branch t_A . Define

$$\Phi(X, Y, \widehat{A}, x) = I_A(U_{\sigma(A)}(\varphi(X, Y, \widehat{A}, x))),$$

and

$$\Psi(A) = \overline{\text{span}}\{\Phi(A, d_n(X)) : n \in \mathbb{N}\},$$

exactly as in [1].

All algebraic and norm estimates are identical to the published proof, so $\Phi_A(X) = \Psi(A)$, $\|\Phi_A\| \leq 1$, and A factors through $\Psi(A)$. It remains only to note the measurability. In the source proof, the only non-Borel input is the map $A \mapsto t_A$, obtained there by Jankov-von Neumann uniformization. Here $A \mapsto t_A = \tau(A)$ is Borel. The displayed formula in the source proof for preimages of open balls under Φ therefore gives Borel, rather than merely $\sigma(\Sigma_1^1)$, preimages. Consequently Φ is Borel. Since the Kuratowski-Ryll-Nardzewski selectors d_n are Borel, the map $A \mapsto \overline{\{\Phi(A, d_n(X)) : n \in \mathbb{N}\}}$ is Borel into $\text{SB}(Z)$. Thus Ψ is Borel as well. $\square$

Consequences for the two corollaries

Corollary 1. *The Borel measurable versions of Corollary 5.12 and Corollary 5.13 of [1] hold.*

Proof. For Corollary 5.12, apply Theorem 1 to the Borel family of identity operators on the given Borel subset of REFL, as in the source proof. The isometric-embedding conclusion and the Borel measurability of the point map follow from the Borel map Φ above.

For Corollary 5.13, the source paper states that the proof is a mix of Braga’s parametrized Zippin theorem [2] and Corollary 5.12, and that the incorrect Borel claim in [2] comes from the same rational-amalgamation uniformization step. Before that final step the assignments are Borel; applying Lemma 1 to the resulting Borel rational-code map supplies a Borel branch selector. Repeating the remaining Kurka amalgamation argument gives a space $Z \in \text{SD}$ with a shrinking basis and Borel maps Ψ and ψ with the asserted properties. $\square$

Novelty and verification notes

Run-index and local parsed-corpus searches for this arXiv id, the named theorems and corollaries, and the main uniformization phrases found no previous packet or apparent later answer. Bounded web searches on 2026-06-14 for exact phrases from Question 6.1, including its theorem/corollary numbering and author names, found no later answer besides the source arXiv page and the earlier Braga paper corrected in the source paper.

A deeper citation check found the published version of the source paper as *Forum of Mathematics, Sigma* 9 (2021), e22, DOI 10.1017/fms.2020.68. Crossref reports no citing works for this DOI, while OpenAlex reports exactly one citing work: Cuth–Dolezal–Doucha–Kurka, *Polish spaces of Banach spaces: Complexity of isometry and isomorphism classes*, DOI 10.1017/s1474748023000440. Text extraction from the published PDF of that citing work shows the source paper only in the bibliography and gives no hits for the specific open-question vocabulary. arXiv API exact-phrase searches for the key labels and question phrases also returned zero results. Semantic Scholar API access was rate-limited during the audit.

This is a deeper bounded novelty check, not an exhaustive proof of historical priority.

The main review point is Lemma 1: once the closed tree is chosen from the Borel graph of η , every downstream step is the published proof with “Borel” substituted for “ $\sigma(\Sigma_1^1)$ ” at the single uniformization bottleneck.

References

- [1] L. Antunes, K. Beanland, and B. de Mendonca Braga, *Uniformly Factoring Weakly Compact operators and Parametrized Dualization*, arXiv:1909.07475, 2019.
- [2] B. M. Braga, *Duality on Banach spaces and a Borel parametrized version of Zippin’s theorem*, Ann. Inst. Fourier (Grenoble) 65 (2015), no. 6, 2413–2435.
- [3] A. S. Kechris, *Classical Descriptive Set Theory*, Graduate Texts in Mathematics 156, Springer, 1995.
- [4] O. Kurka, *Amalgamations of classes of Banach spaces with a monotone basis*, Studia Math. 234 (2016), no. 2, 121–148.

Human Review: Borel Factor-Subspace Selection

Original automatic proof: `solution_packet.pdf`

Checked date: 19 June 2026

Status: Partially manually checked; author verification recommended.

Verdict: The proof looks right, but I am not able to verify every technical detail to the appropriate level.

Human Comments

The proposed argument addresses Question 6.1 of Antunes–Beanland–Braga, *Uniformly factoring weakly compact operators and parametrised dualisation*. The question asks whether the Borel measurable versions of Theorem 5.1, Corollary 5.12, and Corollary 5.13 hold.

The proof appears to make one precise repair to the published proof of Theorem 5.1. In the original argument, after constructing the Borel code map

$$\eta = \psi \circ \sigma : \mathcal{B} \longrightarrow \mathbb{N}^{\mathbb{N}},$$

the authors choose a closed tree whose projection is the analytic set $\eta(\mathcal{B})$. A branch above each code is then chosen using Jankov–von Neumann uniformization, which gives only $\sigma(\Sigma_1^1)$ -measurability. The proposed proof observes that $\eta(\mathcal{B})$ is not an arbitrary analytic set: it is the image of a Borel map. It therefore builds the closed tree from a closed parametrization of the Borel graph of η , obtaining a genuinely Borel branch selector.

The central descriptive-set-theoretic lemma looks correct. Given a Borel map $\eta : S \rightarrow \mathbb{N}^{\mathbb{N}}$, one first codes the standard Borel space S as a Borel subset of Baire space, considers the Borel graph

$$G = \{(i(s), \eta(s)) : s \in S\},$$

and represents this Borel set as a one-to-one continuous image of a closed set. This gives a closed set

$$C \subseteq \mathbb{N}^{\mathbb{N}} \times \mathbb{N}^{\mathbb{N}}$$

and a Borel lift $\tau : S \rightarrow C$ with $p(\tau(s)) = \eta(s)$. This seems to supply exactly the missing Borel branch choice.

The key structural check also appears to work. Every branch of the new tree still has first coordinate equal to $\eta(A_0) = \psi(\sigma(A_0))$ for some original parameter $A_0 \in \mathcal{B}$. Thus the modified tree should not introduce any new “bad” branch spaces. The reflexivity argument in the Kurka amalgamation step is therefore the same as in the source proof: the branch spaces are reflexive, and Kurka’s proposition then gives the reflexivity of the amalgamated space Z .

The measurability repair also appears sound. In the source proof, the proof of Claim 5.11 explicitly reduces the measurability of Φ to the measurability of the assignment $A \mapsto t_A$, together with already Borel objects such as $\sigma(A)$, φ_A , and the fixed selectors. Replacing the old $\sigma(\Sigma_1^1)$ -measurable selector by the Borel selector $t_A = \tau(A)$ should therefore upgrade the same displayed formula from $\sigma(\Sigma_1^1)$ -measurability to Borel measurability. The Borelness of $\Psi(A) = \overline{\text{span}}\{\Phi(A, d_n(X)) : n \in \mathbb{N}\}$ then follows from the usual Kuratowski–Ryll–Nardzewski selectors and the standard Borel closed-span operation.

Citation Correction

There is a small citation-numbering correction in the automatic proof. The rational-code map

$$\psi : \text{mbs} \rightarrow \mathbb{N}^{\mathbb{N}}$$

should be cited as coming from Lemma 5.9 of Antunes–Beanland–Braga, not Lemma 5.10. In the source paper, Lemma 5.9 is the result defining and recording the properties of ψ ; Claim/Lemma 5.10 in that proof is the later reflexivity claim for the interpolation space Z .

Caveat

Although the method appears solid and the single proposed modification seems to target exactly the obstruction in the published proof, I am not sufficiently familiar with all the descriptive-set-theoretic and parametrized Banach-space machinery involved to certify the proof at the level one would want before making a firm mathematical claim. In particular, small technical issues could still be hidden in the passage from the modified closed tree to the exact Kurka amalgamation framework, or in the precise Borel coding of the resulting maps.

Review Outcome

This should be treated as a promising and apparently correct solution, but not as fully human-certified at this stage. I recommend sending the argument to the authors to ask whether the Borel graph/closed-tree replacement indeed removes the $\sigma(\Sigma_1^1)$ -measurability obstacle in Question 6.1.